

(問5.3)

$$\bullet \frac{\partial}{\partial x} (f(x) + g(x)) = \frac{\partial}{\partial x} f(x) + \frac{\partial}{\partial x} g(x) \text{ を示す。}$$

(証明)

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}, A = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix}, B = \begin{bmatrix} b_{11} & \cdots & b_{1n} \\ \vdots & & \vdots \\ b_{n1} & \cdots & b_{nn} \end{bmatrix}$$

$$f(x) = Ax, g(x) = Bx \text{ とする。}$$

$$\begin{aligned} f(x) &= \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \\ &= \begin{bmatrix} a_{11}x_1 + \cdots + a_{1n}x_n \\ \vdots \\ a_{n1}x_1 + \cdots + a_{nn}x_n \end{bmatrix} \end{aligned}$$

$$\begin{aligned} g(x) &= \begin{bmatrix} b_{11} & \cdots & b_{1n} \\ \vdots & & \vdots \\ b_{n1} & \cdots & b_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \\ &= \begin{bmatrix} b_{11}x_1 + \cdots + b_{1n}x_n \\ \vdots \\ b_{n1}x_1 + \cdots + b_{nn}x_n \end{bmatrix} \end{aligned}$$

$$f(x) + g(x) = \begin{bmatrix} (a_{11}+b_{11})x_1 + \cdots + (a_{1n}+b_{1n})x_n \\ \vdots \\ (a_{n1}+b_{n1})x_1 + \cdots + (a_{nn}+b_{nn})x_n \end{bmatrix}$$

$$\frac{\partial}{\partial x}(f(x)+g(x)) = \begin{bmatrix} a_{11}+b_{11} & \cdots & a_{1n}+b_{1n} \\ \vdots & & \vdots \\ a_{n1}+b_{n1} & \cdots & a_{nn}+b_{nn} \end{bmatrix} \quad \text{①}$$

$$\frac{\partial}{\partial x} f(x) = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix}, \quad \frac{\partial}{\partial x} g(x) = \begin{bmatrix} b_{11} & \cdots & b_{1n} \\ \vdots & & \vdots \\ b_{n1} & \cdots & b_{nn} \end{bmatrix}$$

$$\frac{\partial}{\partial x} f(x) + \frac{\partial}{\partial x} g(x) = \begin{bmatrix} a_{11}+b_{11} & \cdots & a_{1n}+b_{1n} \\ \vdots & & \vdots \\ a_{n1}+b_{n1} & \cdots & a_{nn}+b_{nn} \end{bmatrix} \quad \text{②}$$

①, ②より, $\frac{\partial}{\partial x}(f(x)+g(x)) = \frac{\partial}{\partial x} f(x) + \frac{\partial}{\partial x} g(x)$
 が成り立つ (終)

$$\cdot \frac{\partial}{\partial x}(cf(x)) = c \frac{\partial}{\partial x} f(x)$$

(証明)

$$f(x) = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \quad \text{と置く。}$$

$$\begin{aligned} \frac{\partial}{\partial x}(cf(x)) &= \frac{\partial}{\partial x} \left(c \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \right) \\ &= \frac{\partial}{\partial x} \left(\begin{bmatrix} ca_{11}x_1 + \cdots + ca_{1n}x_n \\ \vdots \\ ca_{n1}x_1 + \cdots + ca_{nn}x_n \end{bmatrix} \right) \end{aligned}$$



$$= \begin{bmatrix} ca_{11} & \cdots & ca_{1n} \\ \vdots & & \vdots \\ ca_{n1} & \cdots & ca_{nn} \end{bmatrix} = C \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix} \quad (1)$$

$$\frac{\partial}{\partial x} f(x) = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix}$$

$$c \frac{\partial}{\partial x} f(x) = C \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix} \quad (2)$$

①, ② より $\frac{\partial}{\partial x} (cf(x)) = c \frac{\partial}{\partial x} f(x)$ (終)

• $\frac{\partial}{\partial x} f(cx+lb) = C \frac{\partial}{\partial y} f(y) \quad (y=cx+lb) \in \mathbb{R}^n$

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}, \quad lb = \begin{bmatrix} l_1 \\ \vdots \\ l_n \end{bmatrix}, \quad c = \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}$$

(証明)

$$f(x) = D x \quad \text{ただし } D = \begin{bmatrix} d_{11} & \cdots & d_{1n} \\ \vdots & & \vdots \\ d_{n1} & \cdots & d_{nn} \end{bmatrix}$$

$$\begin{aligned} f(cx+lb) &= D(cx+lb) \\ &= cDx + Dlb \end{aligned}$$

$$= C \begin{bmatrix} d_{11}x_1 + \cdots + d_{1n}x_n \\ \vdots \\ d_{n1}x_1 + \cdots + d_{nn}x_n \end{bmatrix} + \begin{bmatrix} d_{11}l_1 + \cdots + d_{1n}l_n \\ \vdots \\ d_{n1}l_1 + \cdots + d_{nn}l_n \end{bmatrix}$$

$$f(\psi) = f(cx + a)$$

$$= D(cx + a)$$

$$= c \begin{bmatrix} d_{11}x_1 + \dots + d_{1n}x_n \\ \vdots \\ d_{n1}x_1 + \dots + d_{nn}x_n \end{bmatrix} + \begin{bmatrix} d_{11}a_1 + \dots + d_{1n}a_n \\ \vdots \\ d_{n1}a_1 + \dots + d_{nn}a_n \end{bmatrix}$$

$$\frac{\partial}{\partial x} f(cx + b) = c \begin{bmatrix} d_{11} & \dots & d_{1n} \\ \vdots & & \vdots \\ d_{n1} & \dots & d_{nn} \end{bmatrix} \quad \text{①}$$

$$\frac{\partial}{\partial \psi} f(\psi) = \frac{\partial}{\partial (cx + a)} \begin{bmatrix} d_{11}(cx_1 + a_1) + \dots + d_{1n}(cx_n + a_n) \\ \vdots \\ d_{n1}(cx_1 + a_1) + \dots + d_{nn}(cx_n + a_n) \end{bmatrix}$$

$$= \begin{bmatrix} d_{11} & \dots & d_{1n} \\ \vdots & & \vdots \\ d_{n1} & \dots & d_{nn} \end{bmatrix} \quad \text{②}$$

$$\text{①, ② より } \frac{\partial}{\partial x} f(cx + b) = c \frac{\partial}{\partial \psi} f(\psi) \quad (\text{終})$$

(PA 5.4)

$$\begin{aligned} \bullet f(x) &= \max(x_1, x_2, \dots, x_n) \\ &= x_1, x_2, \dots, x_n \end{aligned}$$

$$\frac{\partial}{\partial x} f(x) = \begin{bmatrix} 1 & 0 & \dots & 0 \\ \vdots & 1 & \dots & 0 \\ \vdots & \vdots & & 0 \\ 0 & 0 & \dots & 1 \end{bmatrix} = \underline{E}$$

$$\begin{aligned} \bullet f(x) &= \sqrt{x_1^2 + x_2^2 + \dots + x_n^2} \\ &= (x_1^2 + x_2^2 + \dots + x_n^2)^{\frac{1}{2}} \end{aligned}$$

$$\frac{\partial f(x)}{\partial x} = \frac{1}{\sqrt{x_1^2 + x_2^2 + \dots + x_n^2}}$$

$$\bullet f(x) = |x_1 + x_2 + \dots + x_n|$$

$$(i) \ x_1 + x_2 + \dots + x_n > 0$$

$$f(x) = x_1 + x_2 + \dots + x_n$$

$$\frac{\partial f(x)}{\partial x} = \begin{bmatrix} 1 & 1 & \dots & 1 \end{bmatrix}$$

$$(ii) \ x_1 + x_2 + \dots + x_n < 0$$

$$f(x) = -x_1 - x_2 - \dots - x_n$$

$$\frac{\partial f(x)}{\partial x} = \begin{bmatrix} -1 & -1 & \dots & -1 \end{bmatrix}$$

(P2 5.5)

$$f\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} x_2^2 \\ x_3^2 \\ x_1^2 \end{bmatrix}$$

$$1. \quad \frac{\partial}{\partial x} f(x) = \begin{array}{ccc|ccc} & x_1 & x_2 & x_3 & f(x_1) & f(x_2) & f(x_3) \\ \hline & 0 & 2x_2 & 0 & = x_2^2 & = x_3^2 & = x_1^2 \\ & 0 & 0 & 2x_3 & & & \\ & 2x_1 & 0 & 0 & & & \end{array}$$

$$2. \quad f(f(x)) = f\left(\begin{bmatrix} x_2^2 \\ x_3^2 \\ x_1^2 \end{bmatrix}\right) = \begin{bmatrix} x_3^4 \\ x_1^4 \\ x_2^4 \end{bmatrix}$$

$$3. \quad \frac{\partial}{\partial x} f(f(x)) = \begin{array}{ccc|ccc} & x_1 & x_2 & x_3 & f(x_2^2) & f(x_3^2) & f(x_1^2) \\ \hline & 0 & 0 & 4x_3^3 & = x_3^4 & = x_1^4 & = x_2^4 \\ & 4x_1^3 & 0 & 0 & & & \\ & 0 & 4x_2^3 & 0 & & & \end{array}$$

$$4. \quad \frac{\partial}{\partial x} f(f(x)) = \frac{\partial f(f(x))}{\partial f(x)} \cdot \frac{\partial f(x)}{\partial x}$$

$$\frac{\partial f(f(x))}{\partial f(x)} = \begin{array}{ccc|ccc} & x_2^2 & x_3^2 & x_1^2 & f(x_2^2) & f(x_3^2) & f(x_1^2) \\ \hline & 0 & 2x_3^2 & 0 & = x_3^4 = (x_3^2)^2 & = x_1^4 = (x_1^2)^2 & = x_2^4 = (x_2^2)^2 \\ & 0 & 0 & 2x_1^2 & & & \\ & 2x_2^2 & 0 & 0 & & & \end{array}$$

$$\frac{\partial f(x)}{\partial x} = \begin{bmatrix} 0 & 2x_2 & 0 \\ 0 & 0 & 2x_3 \\ 2x_1 & 0 & 0 \end{bmatrix}$$

$$\frac{\partial}{\partial x} f(f(x)) = \begin{bmatrix} 0 & 2x_3^2 & 0 \\ 0 & 0 & 2x_1^2 \\ 2x_2^2 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 2x_2 & 0 \\ 0 & 0 & 2x_3 \\ 2x_1 & 0 & 0 \end{bmatrix} \\ = \begin{bmatrix} 0 & 0 & 4x_3^3 \\ 4x_1^3 & 0 & 0 \\ 0 & 4x_2^3 & 0 \end{bmatrix} \quad \#$$

(P5.6)
(言証明)

$$A = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix}, \quad B = \begin{bmatrix} b_{11} & \cdots & b_{1n} \\ \vdots & & \vdots \\ b_{n1} & \cdots & b_{nn} \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

$$f(x) = Ax, \quad g(x) = Bx$$

$$f(x) = \begin{bmatrix} a_{11}x_1 + \cdots + a_{1n}x_n \\ \vdots \\ a_{n1}x_1 + \cdots + a_{nn}x_n \end{bmatrix}$$

$$g(f(x)) = \begin{bmatrix} b_{11} & \cdots & b_{1n} \\ \vdots & & \vdots \\ b_{n1} & \cdots & b_{nn} \end{bmatrix} \begin{bmatrix} \sum_{i=1}^n a_{1i}x_i \\ \vdots \\ \sum_{i=1}^n a_{ni}x_i \end{bmatrix}$$

$$= \begin{bmatrix} \sum_{j=1}^n b_{1j} \sum_{i=1}^n a_{ji}x_i \\ \vdots \\ \sum_{j=1}^n b_{nj} \sum_{i=1}^n a_{ji}x_i \end{bmatrix}$$

$$\frac{\partial g(f(x))}{\partial x} = \begin{bmatrix} \sum_{j=1}^n b_{1j} a_{j1} & \cdots & \sum_{j=1}^n b_{1j} a_{jn} \\ \vdots & & \vdots \\ \sum_{j=1}^n b_{nj} a_{j1} & \cdots & \sum_{j=1}^n b_{nj} a_{jn} \end{bmatrix}$$

$$= \begin{bmatrix} (h_{11}a_{11} + h_{12}a_{21} + \dots + h_{1n}a_{n1}) & \dots & (h_{11}a_{1n} + h_{12}a_{2n} + \dots + h_{1n}a_{nn}) \\ \vdots & & \vdots \\ (h_{n1}a_{11} + h_{n2}a_{21} + \dots + h_{nn}a_{n1}) & \dots & (h_{n1}a_{1n} + h_{n2}a_{2n} + \dots + h_{nn}a_{nn}) \end{bmatrix}$$

$$= z'$$

$$BA = \begin{bmatrix} h_{11} & \dots & h_{1n} \\ \vdots & & \vdots \\ h_{n1} & \dots & h_{nn} \end{bmatrix} \begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \dots & a_{nn} \end{bmatrix}$$

$$= \begin{bmatrix} (h_{11}a_{11} + \dots + h_{1n}a_{n1}) & \dots & (h_{11}a_{1n} + \dots + h_{1n}a_{nn}) \\ \vdots & & \vdots \\ (h_{n1}a_{11} + \dots + h_{nn}a_{n1}) & \dots & (h_{n1}a_{1n} + \dots + h_{nn}a_{nn}) \end{bmatrix}$$

$$\therefore \frac{\partial g(f(x))}{\partial x} = BA \quad (\text{終})$$

(問5.7)

$$1. f(g, z) = 2g - 3z$$

(証明)

$$f(g(x), h(x)) = 2g(x) - 3h(x)$$

$$\frac{d}{dx} f(g(x), h(x)) = \frac{d}{dx} (2g(x) - 3h(x))$$

$$= \frac{d}{dx} (2g(x)) - \frac{d}{dx} (3h(x))$$

$$= 2 \frac{d}{dx} g(x) - 3 \frac{d}{dx} h(x)$$

$$= 2g'(x) - 3h'(x) \quad (\text{終})$$

$$2. f(y, z) = yz$$

(証明)

$$\begin{aligned} \frac{d}{dx} f(g(x), h(x)) &= \frac{\partial}{\partial g(x)} f(g(x), h(x)) \frac{\partial}{\partial x} g(x) \\ &\quad + \frac{\partial}{\partial h(x)} f(g(x), h(x)) \frac{\partial}{\partial x} h(x) \\ &= \frac{\partial}{\partial g(x)} g(x) h(x) \frac{\partial}{\partial x} g(x) \\ &\quad + \frac{\partial}{\partial h(x)} g(x) h(x) \frac{\partial}{\partial x} h(x) \\ &= h(x) g'(x) + g(x) h'(x) \\ &= g'(x) h(x) + g(x) h'(x) \quad (\text{終}) \\ &\quad \text{---} \end{aligned}$$

3.

(証明)

$$\begin{aligned} \frac{d}{dx} f(g(x), h(x)) &= \frac{\partial}{\partial g(x)} f(g(x), h(x)) \frac{\partial g(x)}{\partial x} \\ &\quad + \frac{\partial}{\partial h(x)} f(g(x), h(x)) \frac{\partial h(x)}{\partial x} \end{aligned}$$

$$= 2, \quad y = g(x), \quad z = h(x) \text{ と } \exists.$$

$$= \frac{\partial}{\partial y} f(y, z) g'(x) + \frac{\partial}{\partial z} f(y, z) h'(x) \quad (\text{終})$$

$$4. (x(t), y(t)) \rightarrow x(t) = t^2 + t + 1, \quad y(t) = t^2 - t + 1$$

$$E(x, y) = \exp\left(\frac{1}{\sqrt{x^2 + y^2}}\right)$$



$$\frac{\partial}{\partial t} E(x(t), y(t)) = \frac{\partial}{\partial x(t)} E(x(t), y(t)) \frac{\partial x(t)}{\partial t} + \frac{\partial}{\partial y(t)} E(x(t), y(t)) \frac{\partial y(t)}{\partial t}$$

$$\begin{aligned} \frac{\partial}{\partial x(t)} E(x(t), y(t)) &= \frac{\partial}{\partial x(t)} \exp\left(\frac{1}{\sqrt{x(t)^2 + y(t)^2}}\right) (x(t)^2 + y(t)^2)^{-\frac{1}{2}} \\ &= \exp\left(\frac{1}{\sqrt{x(t)^2 + y(t)^2}}\right) \frac{\partial}{\partial x(t)} \left(\frac{1}{\sqrt{x(t)^2 + y(t)^2}}\right) \\ &= \exp\left(\frac{1}{\sqrt{x(t)^2 + y(t)^2}}\right) \left(-\frac{1}{2} (x(t)^2 + y(t)^2)^{-\frac{3}{2}} 2x(t)\right) \\ &= -x(t) (x(t)^2 + y(t)^2)^{-\frac{3}{2}} \exp\left(\frac{1}{\sqrt{x(t)^2 + y(t)^2}}\right) \end{aligned}$$

$$\frac{\partial}{\partial y(t)} E(x(t), y(t)) = -y(t) (x(t)^2 + y(t)^2)^{-\frac{3}{2}} \exp\left(\frac{1}{\sqrt{x(t)^2 + y(t)^2}}\right)$$

$$\frac{\partial x(t)}{\partial t} = 2t + 1, \quad \frac{\partial y(t)}{\partial t} = 2t - 1$$

$$\begin{aligned} (f') &= -x(t) (x(t)^2 + y(t)^2)^{-\frac{3}{2}} \exp\left(\frac{1}{\sqrt{x(t)^2 + y(t)^2}}\right) (2t + 1) \\ &\quad - y(t) (x(t)^2 + y(t)^2)^{-\frac{3}{2}} \exp\left(\frac{1}{\sqrt{x(t)^2 + y(t)^2}}\right) (2t - 1) \end{aligned}$$

$$\begin{aligned} (t^2 + t + 1)(2t + 1) &= 2t^3 + t^2 + 2t^2 + t + 2t + 1 \\ &= 2t^3 + 3t^2 + 3t + 1 \end{aligned}$$

$$\begin{aligned} (t^2 - t + 1)(2t - 1) &= 2t^3 - t^2 - 2t^2 + t + 2t - 1 \\ &= 2t^3 - 3t^2 + 3t - 1 \end{aligned}$$

$$(\vec{z}') = \begin{Bmatrix} -2t^3 - 3t^2 - 3t + 1 - 2t^3 + 3t^2 - 3t + 1 \\ -(2t^3 + 3t^2 + 3t + 1) - (2t^3 - 3t^2 + 3t - 1) \end{Bmatrix}$$

$$(x(t)^2 + y(t)^2)^{-\frac{3}{2}} \exp\left(\frac{1}{\sqrt{x(t)^2 + y(t)^2}}\right) \\ = (-4t^3 - 6t)(x(t)^2 + y(t)^2)^{-\frac{3}{2}} \exp\left(\frac{1}{\sqrt{x(t)^2 + y(t)^2}}\right)$$

$$= (-4t^3 - 6t) \frac{1}{(x(t)^2 + y(t)^2)^{\frac{3}{2}} \sqrt{x(t)^2 + y(t)^2}} \exp\left(\frac{1}{\sqrt{x(t)^2 + y(t)^2}}\right) //$$

(PA 5.8)

$$f\left(\begin{bmatrix} z \\ w \end{bmatrix}\right) = \begin{bmatrix} \frac{e^z}{e^{z+w}} \\ \frac{e^w}{e^{z+w}} \end{bmatrix} \quad y = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$$

$$X = \begin{bmatrix} x_{11} & \dots & x_{1n} \\ x_{21} & \dots & x_{2n} \end{bmatrix}$$

$2 \times n \quad n \times 1$

$$Xy = \begin{bmatrix} x_{11} & \dots & x_{1n} \\ x_{21} & \dots & x_{2n} \end{bmatrix} \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} x_{11}y_1 + \dots + x_{1n}y_n \\ x_{21}y_1 + \dots + x_{2n}y_n \end{bmatrix} \dots z$$

$$\frac{\partial}{\partial x_{ij}} f(Xy) = \begin{bmatrix} \frac{\partial}{\partial x_{ij}} \left(\frac{e^z}{e^{z+w}} \right) \\ \frac{\partial}{\partial x_{ij}} \left(\frac{e^w}{e^{z+w}} \right) \end{bmatrix}$$

$$\frac{\partial}{\partial x_{11}} f(Xy) = \begin{bmatrix} \frac{y_1 e^z (e^{z+w}) - e^z y_1 (e^{z+w})}{(e^{z+w})^2} \\ \frac{(e^{z+w})^2 - e^w y_1 (e^{z+w})}{(e^{z+w})^2} \end{bmatrix} = \begin{bmatrix} 0 \\ \frac{-y_1 e^w}{e^{z+w}} \end{bmatrix}$$

$$\frac{\partial}{\partial x_1} f(x) = \begin{bmatrix} \frac{-e^z y_1 (e^{z+\omega})}{y_1 e^\omega (e^{z+\omega})^2 - e^\omega y_1 (e^{z+\omega})} \\ \frac{y_1 e^\omega (e^{z+\omega})^2 - e^\omega y_1 (e^{z+\omega})}{(e^{z+\omega})^2} \end{bmatrix} = \begin{bmatrix} \frac{-y_1 e^z}{e^{z+\omega}} \\ 0 \end{bmatrix}$$

$$\frac{\partial}{\partial x_{ij}} f(x) = \begin{bmatrix} (1-i) \frac{y_j e^z}{e^{z+\omega}} \\ (-2+i) \frac{y_j e^\omega}{e^{z+\omega}} \end{bmatrix}$$

//

(問6.1)

	表	うS	} \Rightarrow \frac{1}{2}	X... AかBか Y... いった結果
A...	$\frac{1}{4}$	$\frac{3}{4}$		
B...	$\frac{2}{3}$	$\frac{1}{3}$		

$$1. \quad p(X=A, Y=\text{表}) = \frac{1}{2} \times \frac{1}{4} = \frac{1}{8} //$$

$$p(X=A, Y=\text{うS}) = \frac{1}{2} \times \frac{3}{4} = \frac{3}{8} //$$

$$p(X=B, Y=\text{表}) = \frac{1}{2} \times \frac{2}{3} = \frac{1}{3} //$$

$$p(X=B, Y=\text{うS}) = \frac{1}{2} \times \frac{1}{3} = \frac{1}{6} //$$

2.	A	B		$\begin{cases} p(Y=\text{表}) = \frac{11}{24} \\ p(Y=\text{うS}) = \frac{13}{24} \end{cases}$
表	$\frac{1}{8}$	$\frac{1}{3}$	$\frac{11}{24}$	
うS	$\frac{3}{8}$	$\frac{1}{6}$	$\frac{13}{24}$	
	$\frac{1}{2}$	$\frac{1}{2}$		

//

(問 6.2)

$$P(Y=\text{表} | X=A) = \frac{P(X=A, Y=\text{表})}{P(X=A)}$$

$$= \frac{\frac{1}{8} \times \cancel{8^1}}{\frac{1}{8} \times \cancel{8^4}}$$

$$= \frac{1}{4}$$

$$P(Y, X) = P(Y|X)P(X)$$

$$P(X, Y) = P(X|Y)P(Y)$$

(問 6.3)

$$1. P(X=A | Y=\text{表}) = \frac{P(Y=\text{表} | X=A)P(X=A)}{P(Y=\text{表})}$$

$$= \frac{\frac{1}{4} \times \frac{1}{2} \times \cancel{24^3}}{\frac{11}{24} \times \cancel{24}} = \frac{3}{11}$$

$$P(Y=\text{表} | X=B) = \frac{\frac{1}{3} \times \cancel{6^2}}{\frac{1}{2} \times \cancel{6^3}} = \frac{2}{3}$$

$$P(X=B | Y=\text{表}) = \frac{P(Y=\text{表} | X=B)P(X=B)}{P(Y=\text{表})}$$

$$= \frac{\frac{2}{3} \times \frac{1}{2} \times \cancel{24^4}}{\frac{11}{24} \times \cancel{24}} = \frac{8}{11}$$

$$P(X=A | Y=\text{裏}) = \frac{P(Y=\text{裏} | X=A)P(X=A)}{P(Y=\text{裏})}$$

$$= \frac{\frac{3}{4} \times \frac{1}{2} \times \cancel{24^3}}{\frac{13}{24} \times \cancel{24}} = \frac{9}{13}$$

$$P(Y=\text{裏} | X=A) = \frac{\frac{3}{4} \times \cancel{8}}{\frac{1}{2} \times \cancel{8^4}} = \frac{3}{4}$$

$$P(Y=\text{裏} | X=B) = \frac{\frac{1}{3} \times \cancel{6}}{\frac{1}{2} \times \cancel{6^3}} = \frac{1}{3}$$

$$P(X=B | Y=\text{裏}) = \frac{\frac{1}{3} \times \frac{1}{2} \times \cancel{24^4}}{\frac{13}{24} \times \cancel{24}} = \frac{4}{13}$$

2. 表 $\rightarrow$ 表 $A_{\text{表}} = p(\text{表}|A)$ $B_{\text{表}} = p(\text{表}|B)$

$|2| = 1 \times 1$

$$A_1 \text{ 表} \rightarrow A_{\text{表}} = \frac{1}{4} \times \frac{1}{4} = \frac{1}{16}$$

$$B_{\text{表}} \rightarrow B_{\text{表}} = \frac{2}{3} \times \frac{2}{3} = \frac{4}{9}$$

$$A_{\text{表}} \rightarrow B_{\text{表}} = \frac{1}{4} \times \frac{2}{3} = \frac{2}{12} = \frac{1}{6}$$

$$B_{\text{表}} \rightarrow A_{\text{表}} = \frac{2}{3} \times \frac{1}{4} = \frac{2}{12} = \frac{1}{6}$$

$$\frac{9+1+2}{16 \times 9} = \frac{12}{144}$$

$$\frac{7}{9}$$

$$\frac{12}{144} //$$

(P2) 6.5)

$$1. L(\theta) = \prod_{i=1}^N f(x_i; \theta)$$

$$\log L(\theta) = \sum_{i=1}^N \log f(x_i; \theta)$$

$$= \sum_{i=1}^N \log \theta_{x_i}$$

$N = 60$, ① 130 ②, ② ~ ⑥ 130 ②

$$\begin{aligned} \log L(\theta) &= \sum_{i=1}^{30} \log \theta_{x_1} + \sum_{i=1}^6 \log \theta_{x_2} + \sum_{i=1}^6 \log \theta_{x_3} \\ &\quad + \sum_{i=1}^6 \log \theta_{x_4} + \sum_{i=1}^6 \log \theta_{x_5} + \sum_{i=1}^6 \log \theta_{x_6} \\ &= \sum_{i=1}^{30} \log \theta_{x_1} + \sum_{i=1}^6 \log (\theta_{x_3} \theta_{x_4} \theta_{x_5} \theta_{x_6}) // \end{aligned}$$

(PA 6.6)

$$p(X=k) = \frac{\lambda^k e^{-\lambda}}{k!} \quad (k=0,1,2,\dots)$$

λ 平均 = 12 人

$$1. \log L(\lambda) = \sum_{i=1}^{20} \log \left(\frac{\lambda^{(10+i)} e^{-\lambda}}{(10+i)!} \right)$$

$$2. \frac{\partial \log L(\lambda)}{\partial \lambda} = \frac{\partial}{\partial \lambda} \sum_{i=1}^{20} \{ (10+i) \log \lambda - \lambda - \log(10+i)! \}$$
$$= \sum_{i=1}^{20} \left\{ (10+i) \frac{1}{\lambda} - 1 \right\} = 0$$

$$\sum_{i=1}^{20} (10+i) \frac{1}{\lambda} - 20 = 0$$

$$\frac{1}{\lambda} \sum_{i=1}^{20} (10+i) = 20$$

1	2	3	4	5	6	7	8	9	10
20	19	18	17	16	15	14	13	12	11

$$\lambda = \frac{\sum_{i=1}^{20} (10+i)}{20}$$

21	21	21	21	21	21	21	21	21	21
----	----	----	----	----	----	----	----	----	----

$$= \frac{20 \cdot 20 + \sum_{i=1}^{20} i}{20}$$

$$= \frac{410}{20}$$

$$= 20.5$$

$$\lambda = 20.5$$

(問6.7)

$$p(\theta) = \begin{cases} 3 & (\frac{1}{3} \leq \theta \leq \frac{3}{2}) \\ 0 & (\text{otherwise}) \end{cases}$$

21 = 10回振り子 → 9回来, 1回去

$\frac{1}{3} \leq \theta \leq \frac{3}{2}$ のとき,

$$\begin{aligned} P(x_1, \dots, x_{10} | \theta) p(\theta) &= \theta^9 (1-\theta)^1 \times 3 \\ &= 3 \theta^9 (1-\theta)^1 \end{aligned}$$

$$\log P(x_1, \dots, x_{10} | \theta) p(\theta) = \log 3 \theta^9 (1-\theta)$$

$$= \log 3 + 9 \log \theta + \log(1-\theta)$$

$$\begin{aligned} \frac{\partial \log P(x_1, \dots, x_{10} | \theta) p(\theta)}{\partial \theta} &= 9 \frac{1}{\theta} + \frac{-1}{1-\theta} \\ &= \frac{9-9\theta - \theta}{\theta(1-\theta)} \\ &= \frac{9-10\theta}{\theta(1-\theta)} \end{aligned}$$

$$\frac{9-10\theta}{\theta(1-\theta)} = 0 \text{ のとき}$$

$$\frac{1}{3} \leq \theta \leq \frac{3}{2} \text{ かつ } \theta(1-\theta) \neq 0$$

$$9-10\theta = 0$$

$$\theta = \frac{9}{10}$$

(問6.8)

生徒100人

○ A: 20点 = 50人 80点 = 50人

○ B: 0点 = 1人, 1点 = 1人, 2点 = 1人, ..., 99点 = 1人

平均と標準偏差

$$\sqrt{(x-\bar{x})^2 \text{の平均}}$$

$$\begin{array}{r} 5 \overline{)450} \\ 5 \overline{)90} \\ 3 \overline{)18} \\ 3 \overline{)6} \\ 2 \end{array}$$

Aについて

$$\text{平均} \quad \frac{20 \times 50 + 80 \times 50}{100} = 50$$

標準偏差

$$\sqrt{\frac{1}{100} \left(\sum_{i=1}^{50} \frac{(20-50)^2}{900} + \sum_{i=1}^{50} \frac{(80-50)^2}{900} \right)}$$

$$= \sqrt{\frac{1}{100} \times 50 \times 900} = \sqrt{450} = 15\sqrt{2}$$

$\left\{ \begin{array}{l} A \text{ 平均 } 50 \\ \text{標準偏差 } 15\sqrt{2} \end{array} \right.$

Bについて

0 1 2 3 4 5 6 7 8 9 10 ...

99 98 97 96 95 94 93 92 91 90 89 ...

$$99 - 0 + 1 = 100/2 = 50$$

$$\text{平均} \quad \frac{99 \times 50}{100} = 49.5$$

$$\begin{array}{r} 49.5 \\ 2 \overline{)99} \\ 19 \\ 18 \\ 10 \end{array}$$

$$\begin{array}{r} 48.5 \\ 49.5 \\ \hline 48.5 \end{array}$$

標準偏差

$$2 - 1 - 1 = 2$$

$$48.5$$

$\left\{ \begin{array}{l} B \text{ 平均 } 49.5 \\ \text{標準偏差 } 48.5 \end{array} \right.$

